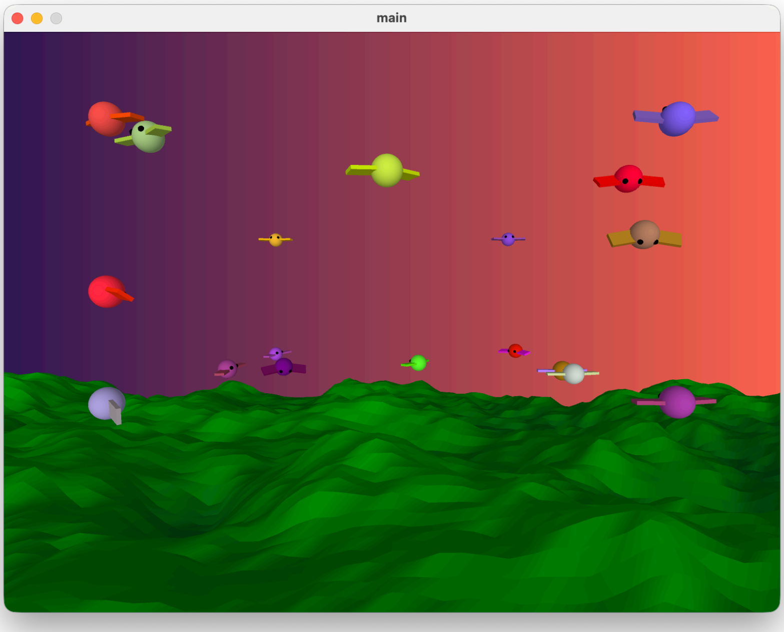
3D Flocking Bird Game

Aaron Struikenkamp | Algorithms (CreaTe) | 14-06-2026

> To find every detail about this project including: process, code, features, sources: https://github.com/AaronMarcusDev/Algorithms-Final/



Description

This is an interactive 3D game where birds flock together. The player can interact with the flock by *throwing rocks* at the birds or *spawning new birds* into the environment. The terrain beneath the birds is generated using Perlin noise and can be regenerated with different height settings through an *in-game menu*.

The bird flocking behavior is driven by the three rules by Craig Reynolds (sourced from Nature of Code in this case). These three steering forces (separation, cohesion and alignment) are calculated per bird every frame and combined to produce natural flock movement.

Birds are spawned using Gaussian distribution. This means most birds appear near the center of the spawn area, with fewer appearing toward the edges, creating a natural clustered starting formation rather than a fully scattered pattern. Each bird has a randomly generated colour and features animated wings that flap continuously using a sine wave based on the system timer, with opposite rotation on each wing.

The player can throw rocks by left-clicking anywhere in the window. When a rock collides with a bird, the bird is removed from the flock and a particle explosion is generated at that location.

Right-clicking spawns a new bird at a random position, allowing the player to grow the flock or replace birds that have been hit. Pressing the 'm' key opens a menu that displays game instructions and allows the player to select the maximum terrain height from preset values. Pressing G regenerates the terrain with the selected height using fresh Perlin noise seeds, creating a completely new landscape each time. Pressing 'r' clears the current flock and respawns the initial seven birds.

The background features a nice gradient with evening colours.
Try it out, have fun! :)

Class Diagram